

AdventureWorks Cycles — Sales & Inventory Analysis

Summary Report | Sales Analytics perspective, with a cross-sectional Operations/Inventory lens

1. Perspective and Focus Area

This analysis takes a Sales Analytics perspective, extended into a cross-sectional view of Operations/Inventory, to answer a connected business question: how is AdventureWorks performing commercially, what is driving that performance, and is finished-goods inventory positioned to support it? The Power BI report is structured as five pages that build on one another: (1) Commercial Performance — revenue, margin, demand segmentation and geography; (2) Sales & Demand Drivers — revenue trend, product and category contribution, pricing and promotion; (3) Inventory & Demand Alignment — whether finished-goods stock is aligned with recent demand; (4) SKU Action Matrix — the specific products warranting review; and (5) an Executive Summary synthesizing findings and recommendations across all four.

2. Key Metrics and Segment Definitions

- **Inventory Attention Value.** The dollar value (stock quantity × standard cost) of finished-goods inventory held in the three attention-worthy status categories below — Zero Stock – Recent Demand, Below Safety Stock, and No Recent Demand. It represents the portion of inventory management should prioritize reviewing, not total inventory value.

Inventory Position Status — each product is assigned to exactly one of the following, checked in this fixed priority order (first match wins):

- **No Inventory Record** — No stock record exists for this product at the Finished Goods Storage location.
- **Zero Stock – Recent Demand** — Zero units on hand, but the product has been ordered within the last 90 days — an active stock-out against live demand; the most urgent status.
- **Below Reorder Point** — On-hand stock has fallen below the product's defined reorder threshold.
- **Below Safety Stock** — On-hand stock has fallen below the minimum safety-stock buffer — a more urgent shortage signal than Below Reorder Point.
- **No Recent Demand** — Stock exists, but the product has not been ordered in the last 90 days.
- **Above Reorder Point** — Stock is healthy relative to the reorder threshold — the default, no-concern status.

Commercial Demand Segment — each product is ranked against all others by Sales Revenue and separately by Total Units Sold, then split at the top-25% mark on each axis:

- **High Demand** — Top 25% by both revenue and units — strong, well-rounded performers across both dimensions.
- **Volume Driven** — Top 25% by units, but not by revenue — sells in high volume at a comparatively lower price point.
- **Value Driven** — Top 25% by revenue, but not by units — a comparatively higher-value, lower-volume product.
- **Lower Demand** — Outside the top 25% on both measures — the long tail of the product portfolio.

Note on Components and Finished Goods Stock

Under the Finished Goods Storage location scope used throughout this analysis, the Components category shows zero recorded stock. This does not mean AdventureWorks holds no Components inventory — it means none of it sits in the finished-goods warehouse; recorded stock instead sits in upstream manufacturing/work-in-process locations (e.g. subassembly, frame forming). As a result, Components is excluded from the Days of Inventory chart (the underlying ratio is blank/undefined with zero stock) and does not appear in finished-goods-scoped inventory KPIs. This is presented as a scope finding warranting further operational review — confirming whether this reflects a build-to-order fulfilment process or a gap in inventory recording — rather than as a stock-out or a data error.

3. Key Assumptions

- **Data period.** The dataset spans 31 May 2011 to 30 June 2014. All rolling-window metrics (e.g. 90-day demand) are anchored to the last recorded sale date in the data (30 June 2014), not the system date, so results remain stable and reproducible. Both 2011 (8 months) and 2014 (6 months) are partial calendar years in the data; year-over-year comparisons on the Yearly Revenue chart should be read on a like-for-like same-period basis, not full-year totals — see Key Insights.
- **Margin basis.** Gross margin uses each product's current/latest Standard Cost applied across all historical sales, rather than reconstructing cost-at-time-of-sale from the cost-history table. This limits model complexity; margin figures should be read as an approximation, not a period-accurate P&L calculation.
- **Inventory scope.** “Finished Goods Stock” is scoped strictly to the Finished Goods Storage location, excluding manufacturing/work-in-process locations — see the Components note above.
- **Promotions.** Promotional activity is identified as any line item with a Special Offer other than “No Discount.”
- **Blank values.** 90-Day Unit Demand and Days of Inventory display as blank for products with zero recorded sales in the trailing 90-day window, since $\text{stock} \div \text{daily demand}$ is mathematically undefined when demand is zero. This is expected model behavior, not missing or erroneous data, and affects a subset of rows on the SKU Action Matrix (Page 4).

4. Methodology and Derivations

Revenue, gross profit and margin are calculated from Sales Order Detail line items, joined through Product → Subcategory → Category. Revenue and Unit Volume Contribution % express each category's share of company-wide totals. 90-Day Unit Demand sums units sold in the 90 days trailing the last recorded order date; Days of Inventory divides total Finished Goods Stock by total 90-day demand (converted to a daily rate) at the aggregate level — a ratio of totals, not an average of per-product ratios, to avoid distortion from SKUs with near-zero recent demand. Revenue by Region is derived by joining Sales Order Header to the Sales Territory table's Regional Group field. Commercial Demand Segment ranks each product against all others on Sales Revenue and, separately, on Total Units Sold (dense rank ÷ total product count, as a percentile), then classifies it into one of four segments based on whether it falls in the top 25% on each axis.

5. Key Insights

- Bikes generates 86% of company revenue but carries a modest 11% gross margin, while Accessories, at just 1.2% of revenue, carries the highest margin in the portfolio (~50%).
- Bikes contributes 86% of revenue from only 33% of unit volume — a high-value, lower-volume category, consistent with its higher average selling price.
- North America accounts for 72% of revenue, with Europe (18%) and Pacific (10%) well behind, indicating a heavily concentrated geographic revenue base.
- On a like-for-like Jan–Jun basis, revenue grew every year: 2014 H1 revenue (\$20.1M) is the highest first-half total on record, ahead of 2013 H1 (\$18.7M) and 2012 H1 (\$17.2M). The apparent full-year decline in the Yearly Revenue chart is a partial-year artifact — the data ends 30 June 2014, not a genuine slowdown — and 2011 is similarly partial (data begins 31 May 2011).
- Promotions touch 13% of units sold but only 7% of revenue, suggesting limited overall pricing impact.
- 49% of finished-goods products (144 of 295) show no recorded stock at the Finished Goods Storage location, driven mainly by Components; of the remainder, \$3.37M in inventory value sits across products flagged Below Safety Stock, No Recent Demand, or Zero Stock with active demand.

6. Recommendations

- Confirm with Operations whether Components' absence from Finished Goods Storage reflects a genuine build-to-order process or a gap in inventory recording — this materially affects how the category's inventory risk should be read.
- Prioritize replenishment review for Bikes SKUs flagged Below Safety Stock first, given the category's outsized share of company revenue.
- Investigate margin-improvement levers for Bikes specifically, given its scale but comparatively thin margin relative to Accessories and Clothing.
- Adopt the SKU Action Matrix (Page 4) as a recurring, filterable review tool for operations and inventory planning meetings rather than a one-time report.